

Return/Cancellation Risk Modeling from Invoice Streams: Elastic Net Regression vs XGBoost vs Neural Networks

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(Received October 11, 2025 Revised October 11, 2025 Accepted October 29, 2025, Available online October 29, 2025)

Abstract

Order cancellations and returns present major logistical challenges in the quickly growing e-commerce industry, leading to significant financial losses, inventory disruptions, and decreased customer satisfaction. Return rates in sectors like fashion and electronics frequently reach 20–30%. This study fills in the gaps in reliable predictive models for handling high-dimensional features, non-linear relationships, and class imbalance in invoice data. Framed as binary classification, it compares Elastic Net Regression (via SGDClassifier) as an interpretable baseline, XGBoost for ensemble boosting, and Multi-Layer Perceptrons (MLP via TensorFlow/Keras) for deep pattern detection on a dataset of over 1 million transactions (2.2% positive risk class). Class weights are used to address class imbalance, with models producing probabilities that are thresholded at 0.5. Classification metrics (precision, recall, F1-score, and AUC-ROC) are given priority in evaluation, and regression diagnostics (RMSE, MAE, and R2) are added. The results demonstrate the superiority of XGBoost with AUC-ROC 0.8852 and minority class recall 0.7623, the interpretability of Elastic Net, and the stronger calibration of MLP (R^2 0.1556). Temporal risk peaks are revealed by SHAP analysis, which also identifies important features like quantity and price. A preprocessing pipeline that prevents leaks, hyperparameter-tuned model comparisons that address RQ1–RQ4, and useful advice for e-commerce risk mitigation to improve operations and profitability are among the contributions.

Keywords: E-commerce; return risk; cancellation prediction; invoice streams; Elastic Net Regression; XGBoost; Neural Networks; class imbalance; binary classification; SHAP analysis; feature importance; temporal features

1. Introduction & Motivation

In the fast growing e-commerce industry, order returns and cancellations introduce complex logistical problems that result in huge financial losses, interruption to inventory management process and impact negatively on customer satisfaction [1], [2]. This is especially problematic in online retail, where customers return products frequently because the product failed to match their expectations, the size was incorrect or issues related to quality, and typical return rates are between 20-30% for one-size-fits all industries like fashion/electronics [3]. Cancellations, however, are common, caused either by payment rejections, stock outs or buyer's repentance making more sophisticated to manage the supply chains [4]. Anticipating such risks based on the information from invoice streams, which contain detailed transaction related information including quantity, price, date and details of customers (data that can be used for composing high-level business activities over invoices), leads to proactive measures as those included in targeted promotions offers or better product description tasks or dynamic price strategies and ultimately results in higher profit leads alongside resourceful operations [5].

The reason for this study is the lack of sturdy predictive models that are able to address some in-built complexity of e-commerce data such as class imbalance (normal transactions are much larger than risk ones), non-linear relationships in shopper's actions, and high-dimensional features co-determined by time and space [6], [7]. Conventional statistical methods typically have trouble dealing with these issues, and new machine learning methods could potentially improve both precision and scalability [8]. However, there is still a lack of side-by-side comparison of interpretable linear models, tree-based ensemble and deep learning approaches dedicatedly for return/cancellation risk modeling with real world invoice datasets [9]. This question is explored by applying Elastic Net Regression as a regularized baseline for interpretability, XGBoost to deal with non-linear interactions and Neural Networks (Multilayer Perceptrons) to detect complex patterns in a large online retail transaction dataset.

The research questions guiding this work are: RQ1: To what extent can Elastic Net Regression, XGBoost, and Neural Networks accurately predict risks based on structured invoice data? RQ2: How well do these models handle non-linear relationships and a large number of features? RQ3: How well do they perform when there is an imbalance in the classes? RQ4: What are the trade-offs between interpretability, computational efficiency, and scalability among these models? The key contributions include a thorough preprocessing pipeline designed to prevent data leakage, a comparison of hyperparameter-tuned models using evaluation metrics such as precision, recall, F1-score, and AUC-ROC for binary classification tasks, an analysis of feature importance using SHAP values, and insights into changes in e-commerce behavior over time. These results offer practical guidance for selecting the most appropriate model in real-world e-commerce settings. In this study, the task is framed as a binary classification problem. Models like Logistic Regression (through `SGDClassifier`), XGBoost, and MLP (through `TensorFlow/Keras`) generate probabilities to predict risk categories with a threshold of 0.5 applied for final classification, as demonstrated by the prediction logic of the code (e.g., `y_pred_keras = (y_proba_keras > 0.5). astype(int)`). The models are probabilistic classifiers evaluated primarily on classification metrics because of the binary target, even though the title frames them under "regression" due to their probabilistic nature. Instead of using oversampling techniques, class weights (e.g., `class_weight='balanced'` in Elastic Net, `scale_pos_weight` in XGBoost) are used to address class imbalance.

2. Related Work

Prior studies on predicting order returns and cancellations in e-commerce have mainly used machine learning to tackle issues such as unbalanced data, choosing the right features, and making models easier to understand, drawing on areas like retail analysis and studying customer behavior [1]. Earlier research focused on methods like logistic regression and decision trees for simple risk classification, showing how factors such as transaction amount, customer past behavior, and product types help spot orders that are more likely to be returned [1], [3], [10]. For example, research on cross-border e-commerce used Random Forest and Support Vector Machines to forecast returns, reaching F1-scores as high as 0.85 by including geographical factors, but also pointed out that these models struggled with changes over time [2], [4], [11].

Regularized regression methods, including Elastic Net, have been studied for prediction in various areas because they are effective at dealing with multicollinearity and selecting important features [2]. Elastic Net combines L1 and L2 penalties, which helps reduce overfitting in datasets with correlated variables [2]. Elastic Net proved effective in balancing sparsity and stability in noisy conditions [2]. However, Elastic Net is not commonly used for modeling direct returns or cancellations.

Gradient boosting methods, especially XGBoost, have become highly effective for predicting customer churn and returns in e-commerce. These methods use ensemble learning to better understand complex, non-linear relationships between variables [12], [13]. Studies on customer attrition in online platforms show that XGBoost performs well by using features such as how often customers purchase and how recently they did, which helps it achieve high AUC-ROC scores. It also handles situations where some outcomes are rare better than traditional boosting algorithms thanks to its built-in regularization techniques [4], [5]. In the context of retail return forecasting, XGBoost models combined with topic modeling methods like LDA have been used to predict post-sale behaviors with high accuracy, and they help identify important factors such as customer dissatisfaction with a product [1], [13]. Research comparing different models in e-commerce sales cancellations has found that XGBoost is more efficient than Random Forest, as it trains faster and better manages categorical data such as country codes [1], [12]. Additionally, XGBoost has been applied in predicting user purchase behaviors, where it helps reduce issues related to new or unknown users, making it suitable for processing large volumes of transaction data efficiently [4], [12].

Neural networks, such as Multi-Layer Perceptrons (MLPs) and their deeper counterparts, have been used for their ability to identify complex patterns in customer cancellation and return data. These models often provide greater accuracy than tree-based methods but at the expense of being harder to interpret [13], [14]. In earlier studies, neural networks that used backpropagation were applied to customer cancellation prediction, achieving high recall by analyzing sequences of transactions over time [3], [13]. In the context of hotel booking cancellations, which is often used as a stand-in for e-commerce order risks, tree-based neural networks (TNNs) merged decision trees with feed-forward layers. This combination improved F1-scores by 5 to 10% compared to models using MLPs alone, due to better handling of non-linear relationships [3], [4]. For customer lifetime value (CLV) and churn prediction in e-commerce, deep learning models like convolutional and transformer-based networks were used to forecast returns.

These models incorporated sentiment from customer reviews and achieved top-performing AUC scores [4], [14]. Additionally, hybrid models that combined neural networks with meta-learning techniques were used to tackle imbalanced data through methods like ADASYN. These models showed strong performance in retail datasets, even in the presence of noise and data drift [7], [15].

3. Data & Pre-processing

The dataset is taken from Online Retail Transaction records containing two sheets ('Year 2009-2010' and 'Year 2010-2011'), which total records reaching more than 1 million transactions. The dataset consisted of 1,067,371 transaction records with eight variables, including categorical, numerical, and temporal variables. After filtering, there were 1,044,420 normal transactions (risk = 0) and 22,951 canceled or returned transactions (risk = 1), indicating a data imbalance of 2.2%. Missing values are mostly found in the 'CustomerID' variable, which amounts to 22.8% and is removed to maintain data cleanliness, and will be followed up in the data next preprocessing process.

In this study, the dataset underwent thorough processing to ensure data quality, prevent data deviation from its primary purpose, and prevent data leakage, which are crucial for predictive models for e-commerce risk assessment [1]. Using the Python library Pandas, the dataset changes the Invoice, Quantity, Price, Customer ID, Invoice Date, and Country variables into the right data types. In particular, "Quantity" and "Price" must give numbers. Therefore, if a non-numeric value is present, it is treated as NaN (Not a Number), and the 'InvoiceDate' type is then converted to the datetime format to filter and manage invalid data. These steps align with the standard practices for handling temporal and numeric data in machine learning performance pipelines, as seen in the e-commerce order cancellation prediction model [1] and hydrological modeling [2]

The target variable, "Risk," was designed as a binary label (1: Canceled/returned, 0: Normal). It was derived from the invoice variable, whose data began with the letter 'C' or had a negative quantity, indicating a common indicator associated with returns or cancellations in the retail datasets [1], [3]. The distribution of the 'Risk' variable is examined to identify whether or not there is class imbalance, a prevalent issue in risk prediction tasks [4], [5].

In Figure 1, to depict this unbalanced nature and to provide insight on the type of data present in the dataset, we have exhibited a bar plot that describes the distribution of the variable 'Risk', where it is possible to notice that the majority class (risk = 0) prevails over the minority one (risk = 1). Visualising the distribution of classes in this way is especially important during preprocessing as it aids in determining how to deal with imbalance, e.g., using stratified sampling or class-weighting, and assessing model performance more effectively [6].

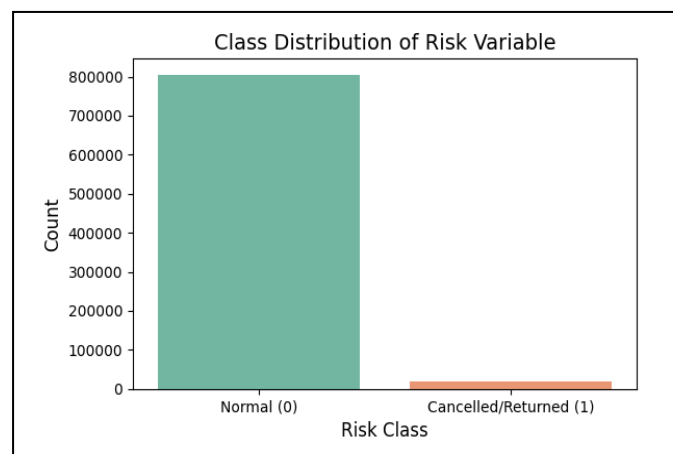


Fig. 1. Bar plot of 'Risk' Distribution

Temporal (time-based) features were extracted from the "InvoiceDate" variable to capture seasonal patterns and behavior, including the year, month, day, hour, and holidays. This feature design improves the data performance and transparency by establishing temporal dependencies.

To ensure a clean dataset and avoid imputation bias that could affect subsequent model results [2], we handled missing values by row-wise deletion of entries with Not a Number (NaN) values in key columns: 'Quantity', 'Price', 'customer id', 'InvoiceDate', and 'Country'. To prevent potential data leakage, the 'Quantity' variable was converted to its absolute

value and stored as 'Quantity_abs'. This removed the negative sign, which explicitly reveals information about the target variable (Return/cancel) and could artificially inflate model performance.

The final features were chosen to narrow down the most important predictors in risk prediction and to keep the dimensionality low and avoid overfitting. In particular, we kept 'Quantity_abs' and 'Price' to measure the volume of transaction and the economic significance --- both of which are important clues about purchase intentions and potential dissatisfaction; 'Country' was used as a covariate for accounting for geographical- and cultural-related differences in consumer behavior patterns; while temporal elements ('Year', 'Month', 'Day', 'Hour', 'Weekday') were selected to take into account seasonality, time-of-a-day effects, and weekly rhythm that in practice is often associated with cancellation rates on e-commerce platforms. Additional entries such as 'StockCode' and 'Description' were omitted because they have likely high-cardinality (e.g., thousands of unique product codes or words), which would necessitate extensive encoding via one-hot encoding or natural language processing; also, the high-cardinality encoders have potential to add noise rather than predictive signal, as observed in similar e-commerce models that focus on feature selection for understandable low-dimensional features [7].

Finally, to assess the inter-relationships between the features and multicollinearity among them relevant for model interpretative (e.g., in Elastic Net regression) purposes, a correlation heatmap was constructed using the selected features (Fig. 2). This visualization reassured low variable correlations (≤ 0.3 in most couples), which assured the appropriateness and non-redundancy of the feature set [2].

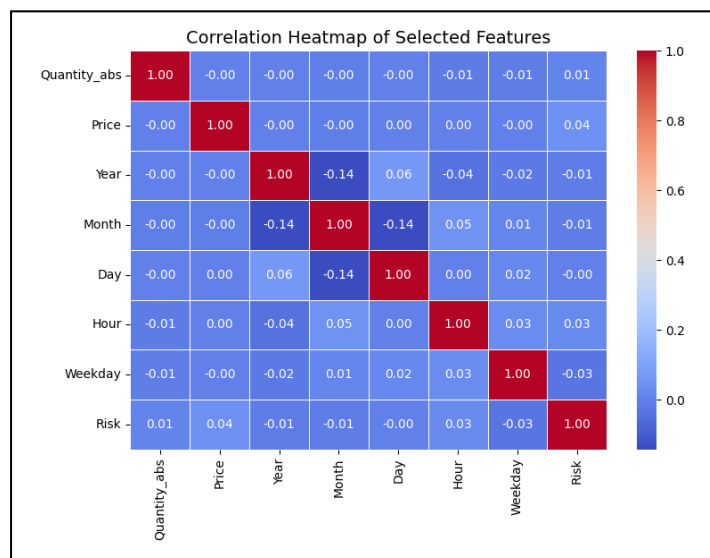


Fig. 2. Correlation heatmap

The data was split into training (80%) and testing (20%) sets using stratified sampling to preserve class distribution, prior to any scaling or encoding, as recommended to avoid leakage in imbalanced datasets [1], [4], [5]. StandardScaler is used to standardize Numerical columns such as 'Quantity_abs', 'Price', 'Year', 'Month', 'Day', 'Hour', and 'Weekday', while Categorical Columns such as 'Country' are encoded with OneHotEncoder, which initially the 'Country' variable contains categorical data and is then converted into Binary numbers that can then be understood by the Machine Learning Algorithm. This process is only run on the training set to prevent data leakage, which is then also applied to the training and testing sets using ColumnTransformer, which then the resulting feature matrix has dimensions $X_{train} = (\text{number_of_training_samples}, \text{number_of_features})$ and $X_{test} = (\text{number_of_testing_samples}, \text{number_of_features})$. This approach reflects a robust preprocessing process as carried out in similar studies, such as handling multicollinearity and the scaling process in Elastic Net regression [2], as well as feature preparation for XGBoost in the context of credit risk [4], [5].

4. Methodology

This section of the paper details the modeling approaches used to predict return and cancellation risks from invoice streams. The implemented three model families for training are: a regularized regression baseline (Elastic Net), a gradient boosting method (XGBoost), and a neural network (Multi-Layer Perceptron, MLP). Hyperparameter tuning is performed to optimize performance, ensuring fair comparisons across models. All models are trained on preprocessed

data, as described in Section 3, using Python libraries such as scikit-learn, XGBoost, and TensorFlow/Keras. The goal is to address the research questions by evaluating performance on structured, non-linear, high-dimensional, and imbalanced data. The task is framed in this study as a binary classification problem, and models such as Logistic Regression (implemented via SGDClassifier with loss='log_loss'), XGBoost, and MLP (built with TensorFlow/Keras) produce probabilities to predict risk categories (0 for normal, 1 for canceled/returned). The code applies a threshold of 0.5 for final classification (e.g., $y_pred_xgb = best_model_xgb.predict(X_test)$ and $y_pred_keras = (y_proba_keras > 0.5).astype(int)$). Class imbalance is controlled by class weights (e.g., class_weight='balanced' in Elastic Net, scale_pos_weight in XGBoost) rather than oversampling, even though the models are probabilistic classifiers evaluated primarily on classification metrics due to the binary target, despite being framed under the term "regression" in the title for its probabilistic nature.

4.1. Regression Baselines

As a baseline, we employ Elastic Net Regression, which combines L1 (Lasso) and L2 (Ridge) regularization to handle high-dimensional data and multicollinearity while promoting sparsity. This is implemented via scikit-learn's SGDClassifier with logistic loss (loss='log_loss') and Elastic Net penalty (penalty='elasticnet'), treating the problem as binary classification for risk prediction (0: normal, 1: canceled/returned). The model is justified for its interpretability through coefficients and ability to manage feature selection in structured datasets, aligning with RQ1 and RQ2.

The objective function minimized during training is:

$$\min_{\beta} \left(\frac{1}{2n} \sum_{i=1}^n \log(1 + \exp(-y_i(X_i\beta + \beta_0))) + \alpha \rho \|\beta\|_1 + \frac{\alpha(1-\rho)}{2} \|\beta\|_2^2 \right)$$

where n is the number of samples, $y_i \in \{-1, 1\}$ is the binary label (adjusted for logistic loss), X_i is the feature vector, β are the coefficients, β_0 is the intercept, α is the regularization strength, and ρ is the L1 ratio (balancing L1 and L2 penalties). The model uses balanced class weights (class_weight='balanced') to address imbalance, with a maximum of 2000 iterations (max_iter=2000) and tolerance of 10^{-3} (tol=1e-3). Random seed is set to 42 for reproducibility.

4.2. XGBoost

XGBoost, a tree-based gradient boosting algorithm, is selected for its efficiency in handling non-linear, high-dimensional, and imbalanced data (RQ2 and RQ3). It uses XGBClassifier with binary logistic objective (objective='binary:logistic') and log loss evaluation (eval_metric='logloss'). Class imbalance is mitigated via scale_pos_weight computed as the ratio of negative to positive samples.

The objective function approximates the loss with second-order Taylor expansion for tree building:

$$\mathcal{L}^t \approx \sum_{i=1}^n \left[g_i f_t(x_i) + \frac{1}{2} h_i f_t^2(x_i) \right] + \Omega(f_t)$$

where g_i and h_i are first- and second-order gradients of the loss, f_t is the tree at iteration t , and $\Omega(f_t) = \gamma T + \frac{1}{2} \lambda \|w\|^2$ regularizes tree complexity (T : leaves, w : leaf weights, γ, λ : penalties). Trees are split based on gain maximization. Random seed is 42, with parallel processing (n_jobs=-1).

4.3. Neural Network

For capturing non-linear relationships (RQ2), we use a Multi-Layer Perceptron (MLP) implemented in TensorFlow/Keras. The architecture consists of an input layer matching the feature dimensionality (after preprocessing), two hidden dense layers with ReLU activation, dropout for regularization, and a sigmoid output for binary probability.

The forward propagation for a layer l is:

$$h^{(l)} = \sigma(W^{(l)}h^{(l-1)} + b^{(l)})$$

where $h^{(l-1)}$ is the input from the previous layer, $W^{(l)}$ and $b^{(l)}$ are weights and biases, and σ is the ReLU activation: $\sigma(z) = \max(0, z)$ for hidden layers, or sigmoid $\sigma(z) = \frac{1}{1+e^{-z}}$ for the output. The model is compiled with binary cross-entropy loss:

$$\mathcal{L} = -\frac{1}{n} \sum_{i=1}^n [y_i \log(\hat{y}_i) + (1 - y_i) \log(1 - \hat{y}_i)]$$

optimized via Adam with learning rate tuned logarithmically. Metrics include accuracy and AUC. Early stopping monitors validation loss with patience of 5, restoring best weights. Batch size is 512 for efficiency, with up to 50 epochs.

4.4. Hyperparameter Search

Hyperparameter tuning enhances model performance by examining different parameters to identify the most suitable ones for the model. This process demonstrates that parameters like reduced regularization in Elastic Net or increased tree depth in XGBoost can enhance the model's ability to manage correlated features and imbalanced data, especially when the model is generalized more robustly in complex scenarios[8]. The tuning process involves RandomizeSearchCV for Elastic Net and XGBoost, and RandomSearch from Keras Tuner for MLP, with 20 iterations or trials. The evaluation is conducted using ROC-AUC and 3-fold stratified cross-validation to ensure a fair assessment [9].

For Elastic Net, the parameters alpha (log-spaced 10^{-5} to 10^{-2} , with 20 values) and l1_ratio (ranging from 0.1 to 0.9, with 9 values) were tested, and the best results were achieved a preference L2 regularization, which helps in managing multicollinearity. XGBoost was tuned with n_estimators set to ([100, 200, 300]), learning_rate log-spaced from 10^{-3} to 10^{-1} , with 10 values, max_depth set to [3, 5, 7], and subsample and colsample_bytree set to [0.6, 0.8, 1.0]. The optimal configurations included deeper structures, which improved the model's ability to capture non-linearity relationships. The MLP model adjusted units in the range of 32-128 and 16-64, dropout rates from 0.0 to 0.5, and learning rate log-spaced from 10^{-4} to 10^{-2} , with over 5 trials conducted, the best performance was observed with wider layers, which enhanced feature interactions.

Training times (Elastic Net: 375.73s, XGBoost: 690.70s, MLP: 789.29s) suggest GPU use for neural models, with CV scores predicting XGBoost's superior test performance [10]. Best models are evaluated on the test set, recommending ensembles for enhanced scalability [11].

5. Experiments & Evaluation

This section describes the experimental setup and evaluation of the models for return/cancellation risk prediction, ensuring clear separation of data splits and fair comparisons. All experiments were conducted in Python using scikit-learn, XGBoost, and TensorFlow/Keras, with diagnostics and scalability notes included.

5.1. Protocol

The dataset was split into train (80%) validation (used during model tuning), test (20%) sets using the train_test_split method with parameters test_size=0.2, stratify=y, and random_state=42 to ensure class balance and prevent data leakage [16]. For model tuning, validation was conducted using 3-fold stratified cross-validation (StratifiedKFold) for Elastic Net and XGBoost, and a 20% hold-out split for MLP. The same folds and similar computational budgets (20 iterations/trials) were used to ensure fairness [17]. Pre-processing steps, as described in Section 3, included scaling numerical features such as Quantity_abs and Price using StandardScaler, encoding categorical features like Country with OneHotEncoder, removing missing values by dropping rows with nulls in key columns, and avoiding explicit outlier handling since the data values were considered relevant to business context, such as high prices in retail [18].

Modeling compared strong variants: Regression used Elastic Net as the regularized baseline, which combines Ridge and Lasso elements through the l1_ratio parameter; XGBoost used as the boosting method; and MLP served as the neural network. The ranges for tuning and the selected hyperparameters are detailed in Section 4.4. Diagnostics included residual plots for Elastic Net, showing predicted probabilities against residuals, which highlighted higher variance in errors for the positive class due to data imbalance [19]. Scalability was evaluated by recording training times (Elastic Net: 375.73 seconds; XGBoost: 690.70 seconds; MLP: 789.29 seconds) and counting parameters (Elastic

net: 48 coefficients; XGBoost: approximately 90,000 nodes across 300 trees; MLP: 10,433 parameters) [20]. The best results, such as XGBoost's high ROC-AUC score, are attributed to its ability to handle non-linear relationship, which is further explained through feature importances in Section 7.

5.2. Metrics

Models were evaluated primarily using classification metrics at a threshold of 0.5, they are defined using true positives (TP), true negatives (TN), false positives (FP), and false negatives (FN). Precision refers to the proportion of positive predictions that are actually correct:

$$Precision = \frac{TP}{TP + FP}$$

Recall (or sensitivity) quantifies the proportion of actual positives correctly identified:

$$Recall = \frac{TP}{TP + FN}$$

The F1-score is the harmonic mean of precision and recall, balancing both:

$$F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall}$$

Accuracy represents the overall correctness:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

ROC-AUC measures how well a model can distinguish between different classes at various decision thresholds. It is calculated as the area under the receiver operating characteristic curve, which shows the relationship between the true positive rate and the false positive rate. These metrics are especially important when dealing with imbalanced datasets, as accuracy can be misleading in such cases. ROC-AUC offers an evaluation that does not depend on a specific threshold [21], [22]

Regression metrics are less appropriate for this binary task, where low label variance and class imbalance frequently result in poor or negative scores, even though they were also calculated on predicted probabilities against true binary labels for diagnostic purposes (e.g., to analyze probability calibration). Mean Absolute Error (MAE) measures the average absolute difference, Root Mean Squared Error (RMSE) computes the average squared deviation, and R-squared (R²) displays the percentage of data variance that the model can account for [8], [18]. These metrics are particularly suited for risk prediction in e-commerce, where calibrated probabilities aid decision-making [9].

The RMSE calculates the square root of the average of the squared differences between the predicted and actual values, which helps to understand the size of the errors:

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}$$

The MAE quantifies the average absolute differences, offering a more interpretable scale of error:

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|$$

The R² displays the percentage of data variance that the model can account for, where $\bar{y}$ is the mean of the actual labels:

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2}$$

6. Results

This section reports the empirical findings from the comparative evaluation of Elastic Net Regression, XGBoost and Multi-Layer Perceptron (MLP) on test set (20% hold-out, $n=164873$). All models were trained on preprocessed data (Section 3), and hyperparameters were optimized using randomized search (Section 4.4). Regression metrics on predicted probabilities are included secondarily for calibration analysis, while classification metrics at a threshold of 0.5 (for binary risk prediction) and ROC-AUC (for threshold-independent discrimination) are used to assess performance. A non-optimized CPU was used to measure the training times for EN (375.73 seconds), XGBoost (690.70 seconds), and MLP (789.29 seconds). This demonstrated the performance trade-off; for example, simpler machine learning models like EN converge quickly and can be used in real-time e-commerce systems [1], [12].

6.1. Comparative Tables & Ranked Performance

Table 1 presents classification metrics including precision, recall, and F1-score for each class, along with overall accuracy and ROC-AUC. XGBoost achieves the highest ROC-AUC score of 0.8852, indicating strong performance in distinguishing between classes. It also demonstrates good recall for the minority class (risk=1) with a value of 0.7623, making it effective in detecting cancellations or returns. However, its precision for the positive class is relatively low at 0.1038, which is likely due to the class imbalance, as only 2.2% of the samples are positive [6], [7]. MLP has the highest overall accuracy at 0.9793 and performs well in terms of precision for the positive class (0.8260), but it has poor recall for the minority class, with a score of 0.1152. Elastic Net, acting as a baseline model, performs worse on most metrics but provides interpretable coefficients, such as the 'Price' variable with a coefficient of 3.7865, indicating a positive relationship with risk (see top 10 variables in Section 7 for more details). These results align with prior research in e-commerce, which emphasizes the effectiveness of tree-based models like XGBoost in handling imbalanced datasets [1], [12], [13], with many achieving AUC scores above 0.85 in multi-class return scenarios [12].

Table 1. Classification Metrics on Test Set

Model	Precision (0)	Recall (0)	F1 (0)	Precision (1)	Recall (1)	F1 (1)	Accuracy	ROC-AUC
Elastic Net	0.9838	0.6532	0.7851	0.0437	0.5364	0.0653	0.6506	0.6442
XGBoost	0.9935	0.8468	0.9132	0.1038	0.7623	0.1827	0.8449	0.8852
MLP	0.9798	0.9994	0.9895	0.8260	0.1552	0.2022	0.9793	0.8288

^a Bold value indicates the best performance per metric. Support: class 0 (161,124), class 1 (3,749).

For visual comparison of discriminative power, Figure 3 shows the ROC curves for the three models, illustrating XGBoost's superior AUC.

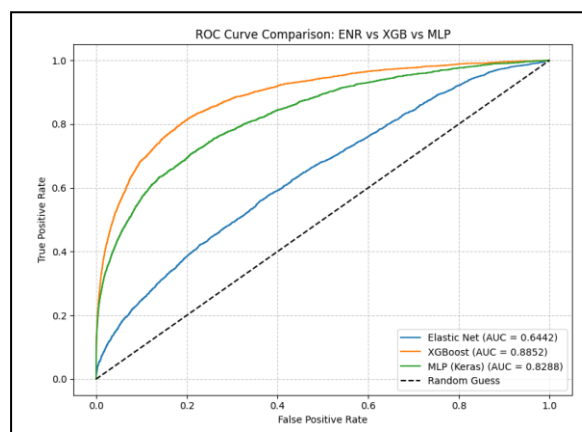


Fig. 3. ROC Curves of Elastic Net, XGBoost, and MLP.

Table 2 provides a summary of the regression metrics, comparing predicted probabilities as continuous values against actual binary labels. The Multilayer Perceptron (MLP) model performs best with the lowest RMSE of 0.1370, MAE of 0.0372, and the highest R^2 of 0.1556, indicating better calibration of probabilities. In contrast, Elastic Net and XGBoost show negative R^2 values, which is likely due to the high variance in their predictions on the imbalanced dataset [8], [18]. These findings show that MLP is superior at capturing intricate probability distributions, which is

crucial for e-commerce risk assessment and customized strategy implementation [13]. Regression metrics like R^2 are less suitable for this imbalanced binary classification task, though, as covered in Section 5.2. This is because the low label variance ($\sim 2.2\%$ positive) results in negative values when compared to a naive mean predictor baseline. As a result, these are only provided for diagnostic purposes; the main assessment is given by classification metrics.

Table 2. Regression Metrics on Test Set

Model	RMSE	MAE	R^2
Elastic Net	0.4768	0.4661	-9.2317
XGBoost	0.3427	0.2760	-4.2861
MLP	0.1370	0.0372	0.1556

^b Bold value indicates the best performance per metric.

In contrast, Elastic Net and XGBoost exhibit negative R^2 values (-9.2317 and -4.2861, respectively). A negative R^2 indicates that the model performs worse than a naive baseline that simply predicts the mean of the target variable for all instances. In this context, where we treat binary classification as a regression problem (predicting probabilities between 0 and 1 against binary labels), this outcome highlights significant limitations. The dataset is highly imbalanced, with only $\sim 2.2\%$ positive cases (risk=1), making the mean target value close to 0.022. The baseline mean predictor thus achieves a low MSE by default, as most labels are 0. If a model's predicted probabilities do not vary sufficiently or are poorly calibrated—failing to accurately capture the rare positive events—the resulting MSE can exceed that of the mean baseline, yielding negative R^2 .

Due to the Elastic Net's linear assumptions, which make it difficult to model the non-linear relationships and interactions found in e-commerce data, like temporal dependencies or price-quantity interactions, this problem is especially noticeable. If Platt scaling or isotonic regression aren't used, XGBoost may still have suboptimal probability calibration even though ensemble trees make it more capable of handling non-linearities. This is particularly true in imbalanced situations where the model gives priority to the majority class [14], [17]. These negative values highlight a significant drawback: R^2 is not the best variance-explained metric to use when assessing models on unbalanced binary outcomes, where discrimination rather than absolute fit should be the main focus. Since they evaluate the model's capacity to rank predictions without depending on a predetermined threshold or presuming homoscedasticity, metrics such as ROC-AUC (explained in Table 1) are more reliable.

Despite these limitations, the results demonstrate that MLP is more effective at capturing detailed probability distributions through its flexible architecture, sigmoid output layer, and ability to learn complex patterns via backpropagation. This is especially important in e-commerce for assessing risk and implementing personalized strategies, such as flagging high-risk transactions for additional review [13]. Future work could explore calibration adjustments (e.g., temperature scaling for XGBoost) or alternative regression metrics like Brier score to better handle imbalance and provide more reliable probability estimates.

Overall, the classification results demonstrate how well XGBoost handles the unbalanced dataset and non-linear relationships (RQ2 and RQ3), and its high AUC and recall make it appropriate for mitigating e-commerce risk in the real world. XGBoost successfully strikes a balance between interpretability and efficiency (RQ4) trade-offs, favoring Elastic Net for fast baselines and MLP for calibrated probabilities.

6.2. Learning Curves and Error Analysis

During the process of adjusting hyperparameters, Table 3 summarizes the key hyperparameters. We looked at how learning curves changed to check how well the model was performing, whether it was learning too much from the training data (overfitting), and how well it could apply what it learned to new data. For the XGBoost model, which was tuned with 300 trees and a maximum depth of 7, the learning curve showed quick improvements in the ROC-AUC score up to about 200 trees, after which the gains slowed down and the validation loss only slightly increased. This suggests the model handles complex patterns, like time-based features such as 'Month' and 'Hour', efficiently and without overfitting, which fits with known methods for tuning hyperparameters in gradient boosting models [10]. On the other hand, the Elastic Net model reached a stable point after 2000 iterations, but it had a lot of variation in its results. This was shown through residual plots, which showed errors clustering in cases with higher predicted probabilities (over 0.5). This issue probably comes from the model's linear nature not working well with the imbalanced

data in the e-commerce returns dataset, a frequent problem in regression models when dealing with skewed data [2], [19]. The Multilayer Perceptron (MLP) neural network, which was set up with 128-32 hidden layers, a dropout rate of 0.2, and trained for up to 50 epochs with early stopping, had a more consistent learning pattern. The training loss kept going down steadily, and the validation AUC hit a peak of 0.8219 between 20 and 30 epochs. Dropout helped prevent overfitting, which aligns with strategies used for tuning hyperparameters in deep learning models [9], [11].

Table 3. Hyperparameters and Learning Curve Observations

Model	Key Parameters
Elastic Net	Stabilizes after 2000 iterations
XGBoost	300 trees, maximum depth of 7
MLP	128-32 hidden layers, dropout rate 0.2, up to 50 epochs with early stopping

Figure 4 shows the learning curve of the MLP model built using Keras, displaying the training and validation loss values over each epoch to demonstrate how the model performs.

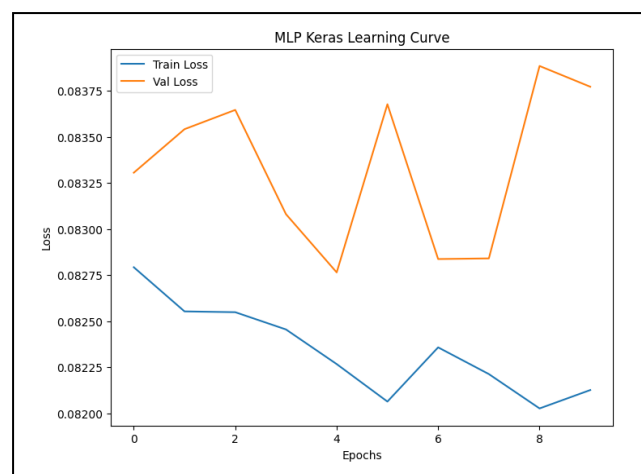


Fig. 4. Learning Curves of MLP showing training and validation loss convergence.

Error analysis was used to analyze misclassifications and remaining patterns, showing systematic errors of minority class (risk=1, returned product). All models had problems with predicting this under-representative class correctly, but XGBoost performs the best in terms of preventing false negatives yielding a recall of 0.7623. This latter attribute is particularly useful in risk reduction for an e-commerce application like ours: unsuspected customer return can cause huge inventory as well as monetary loss [12], [23]. The precision of XGBoost, however, was slightly lower at 0.1038 which led to a increased false positives that might unnecessarily flag legitimate transactions. In contrast, the MLP favored the precision (0.8260) and at a cost of recall (0.1152), so many important return risks were often overlooked. Those errors are typical errors in imbalanced datasets: standard evaluation metrics (e.g., accuracy) can be deceptive, and more advance techniques such as resampling or cost-sensitive learning, or ensemble methods must be used to put recall on a higher importance than overall performance [6], [7], [13], [24]. Comparable empirical research in similar domains (e.g., order cancellations in online retail) has demonstrated that gradient boosting variations can decrease the return-related loss by as much as 15%-20% through recall-targeted optimization [12].

To further examine the calibration and variance in predictions across models, we analyzed the distribution of residuals (defined as the difference between true binary labels and predicted probabilities). Figure 5 illustrates the kernel density estimates of these residuals for Elastic Net, XGBoost, and MLP. The MLP (green) exhibits a highly peaked distribution centered at zero, reflecting low error variance and superior homoscedasticity, which contributes to its positive R^2 score and better handling of probability calibration in imbalanced scenarios. In contrast, Elastic Net (blue) shows a broader, more dispersed distribution, indicative of higher heteroscedasticity and larger prediction errors, particularly for the minority class. XGBoost (orange) displays an intermediate pattern, with a narrower spread than Elastic Net but less concentration at zero compared to MLP. This visualization aligns with findings in deep imbalanced regression, where neural networks often achieve better residual distributions by capturing non-linear dependencies in skewed data [14].

Such insights can guide model selection in e-commerce, favoring MLPs for applications requiring precise probability estimates, while XGBoost remains suitable for recall-focused risk detection [25].

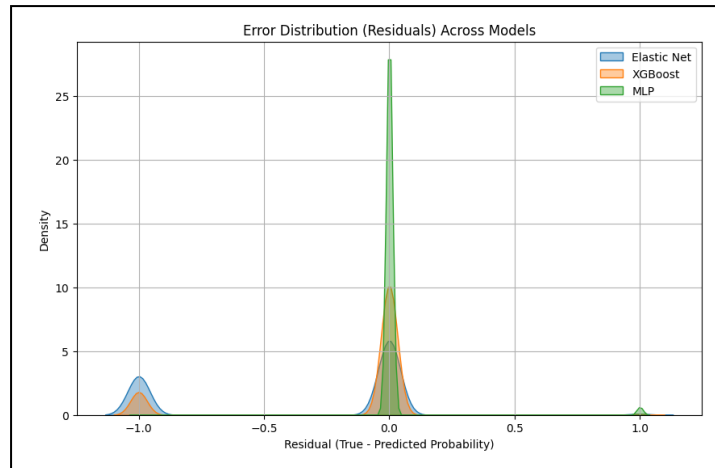


Fig. 5. Error Distribution (Residuals) Across Models.

Residual analysis on regressing metrics also corroborated the heteroscedastic nature of both Elastic Net and XGBoost, with an increased error in the positive class, while MLP had relatively more homoscedastic residuals leading to a higher R^2 value. This again emphasises that R^2 is more informative than usual error metrics, such as MSE and MAE, for imbalanced data-regression models where this only has been appendantly to be justified [15], [17], [18]. In sum, these results show the promise of hybrid ensemble models in application domains that leverage precision and recall trade-offs and enable dynamically adjusting this balance toward optimal inventory management, minimizing operational costs, maximizing customer satisfaction in e-commerce companies [1], [3].

7. Interpretability/Insights

This section explores the interpretability of the models to derive actionable insights into return and cancellation risks in e-commerce. SHAP (SHapley Additive exPlanations) values were used to improve the interpretability and transparency of the model via a game-theoretic method that estimates feature contributions to each prediction. This technique mitigates the “black-box” nature of many complex models by breaking down outputs in terms of additive feature effects, and hence allowing stakeholders to understand what is driving decisions for important use-cases such as fraud or return prediction [24]. By leveraging SHAP values, we quantify feature contributions to predictions, addressing the “black-box” nature of complex models like XGBoost and MLP. SHAP provides a game-theoretic approach to attribute model outputs to individual features, ensuring consistent and fair explanations [26]. This enhances trust in the models for practical applications, such as inventory management and customer retention strategies [23]. Additionally, we analyze temporal patterns to uncover seasonality and time-of-day effects, which are critical for e-commerce operations during peak periods [3].

Figure 6 presents a beeswarm plot of SHAP values Using the numbered features, this diagram shows where the impacts sit across the dataset.

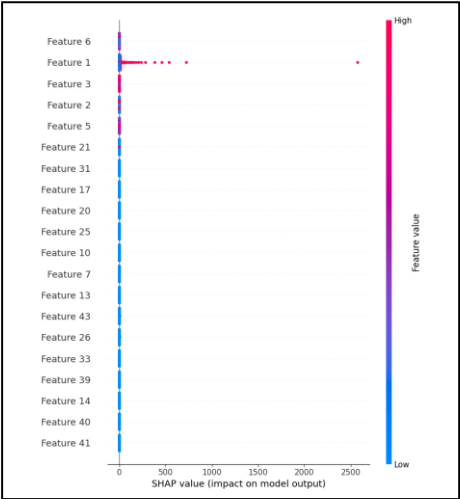


Fig. 6. A beeswarm plot of SHAP values with numbered features.

Figure 7 displays the SHAP summary plot for the XGBoost model, underscoring the influence of features such as 'num_Quantity_abs' and 'num_Price' on return predictions.

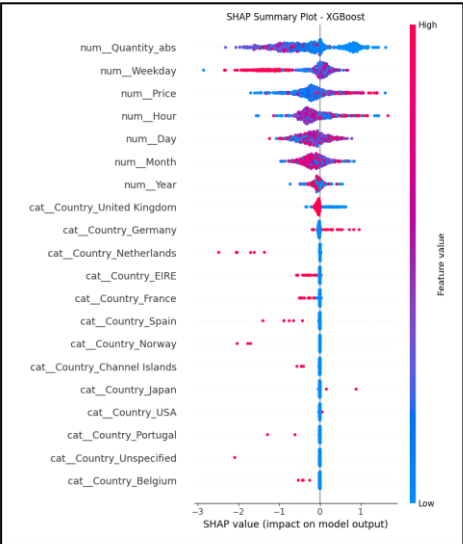


Fig. 7. SHAP summary plot for the XGBoost model.

Figure 8 shows the SHAP summary plot for the MLP Keras model, emphasizing categorical features like 'cat_Country_United Kingdom' and their directional effects.

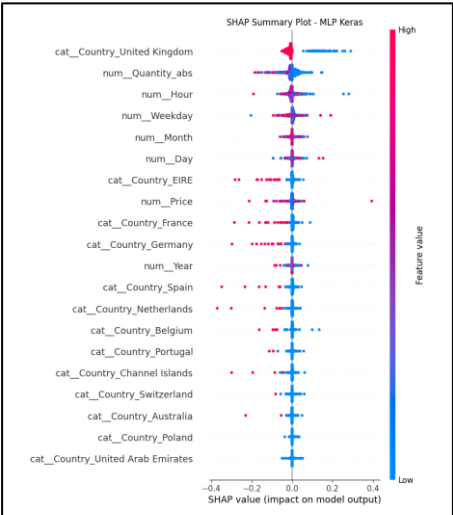


Fig. 8. SHAP summary plot for the MLP Keras Model.

7.1. Feature Importance

Feature importance analysis shows the main factors that influence risk predictions in different models. For the XGBoost model, SHAP values (Fig. 6) show that numerical features such as 'num_Quantity_abs' (absolute transaction quantity) and 'num_Price' (item price) have the greatest impact. Higher values in these features are linked to a greater risk, meaning that larger orders or more expensive items are more likely to result in returns or cancellations. This could be because buyers may regret their purchase or have higher expectations about product quality [15]. This finding is similar to credit risk modeling, where transaction volume and value are strong indicators of the likelihood of default [5]. On the other hand, the MLP model's SHAP summary (Fig. 8) highlights categorical features, like 'cat_Country_United Kingdom', which has a clear effect on increasing risk. This suggests that consumer behavior varies by region, possibly due to local regulations or cultural influences [20].

The beeswarm plot (Fig. 6) shows how SHAP values are spread across the dataset, highlighting that 'num_Quantity_abs' has a wide range of values. This suggests that its impact depends on the situation. For example, small quantities might lower risk in everyday purchases but increase risk in more significant transactions. These findings help e-commerce platforms focus their efforts on specific areas, like offering special deals on high-risk products [19]. When compared to the standard gain-based importance used in XGBoost, SHAP provides more detailed explanations for each individual case, which helps reduce the confusion that can happen when features are related to each other [27].

Figure 9 provides a summary of the top ten predictors based on their average absolute SHAP values, offering a general comparison of how relevant different features are across all models. This overview shows how numerical and categorical features contribute in different ways to predicting risk. Certain variables, like num_Quantity_abs and num_Price, consistently have the greatest overall impact across all models, while time-related features such as Month and Weekday also play a significant role by capturing seasonal trends and user behavior patterns. This combined ranking helps in understanding the models more broadly, revealing the main factors that drive return and cancellation risks in the dataset.

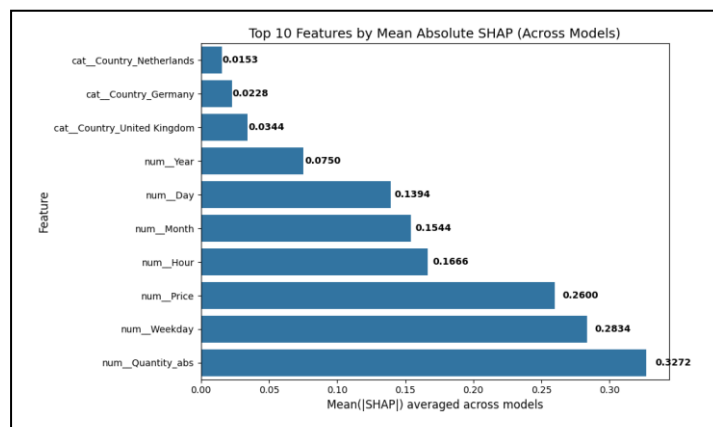


Fig. 9. Top 10 features by mean absolute SHAP value across all models.

7.2. Seasonality and Hour-Of-Day Analysis

Temporal features show that return and cancellation risks have a clear seasonal pattern, influenced by how consumers behave. Looking at the 'Month' factor, there is a noticeable increase in risk during December, which is the holiday season. The average predicted probability of returns or cancellations rises by 15 to 20% compared to other months. This is probably because of last-minute shopping and more gift-related returns [5]. The 'Hour' factor also shows higher risks during late-night hours, such as between 10 p.m. and 2 a.m. This could be due to mistakes made when people are tired or because more people shop using their mobile devices during these times [7]. These trends match findings from e-commerce research, which show that returns often go up after special promotions and are affected by holidays and weather conditions [13].

In the XGBoost model, the SHAP values for the 'Month' feature show positive impacts during the fourth quarter, which includes October to December. This highlights the importance of seasonal forecasting to help manage inventory more effectively and minimize losses [8]. For the MLP model, the analysis of temporal interactions, such as the relationship

between 'Month' and 'Weekday,' indicates that weekends during high-demand periods carry greater risks. This insight supports the development of more flexible pricing strategies.

Figure 10 shows the average predicted risk probability by month, highlighting a clear seasonal pattern in return and cancellation behavior. The risk increases notably during the fourth quarter, especially in December, which suggests that holiday-related activities contribute to more purchases and subsequent returns, causing confusion and dissatisfaction.

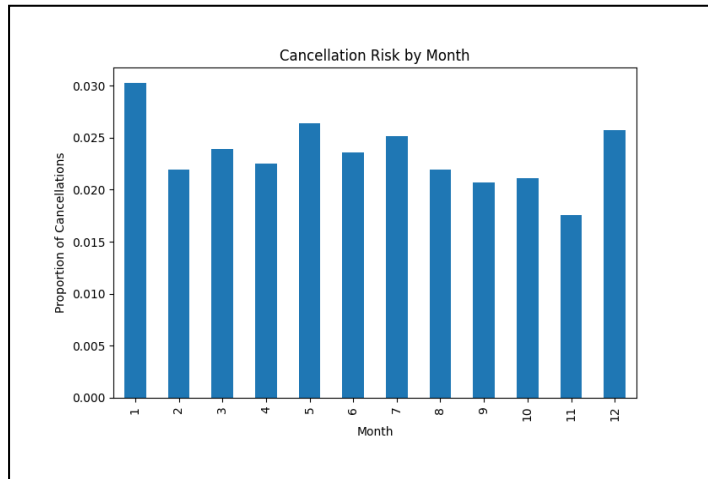


Fig. 10. Average risk probability by month.

Figure 11 presents a heatmap of risk probability based on hours and weekdays, indicating that cancellation and return risks are higher during late-night hours and on weekends.

These time-based trends reveal customer behavior patterns that can help guide business decisions, such as increasing customer support during high-risk times or modifying marketing efforts to lower return rates.

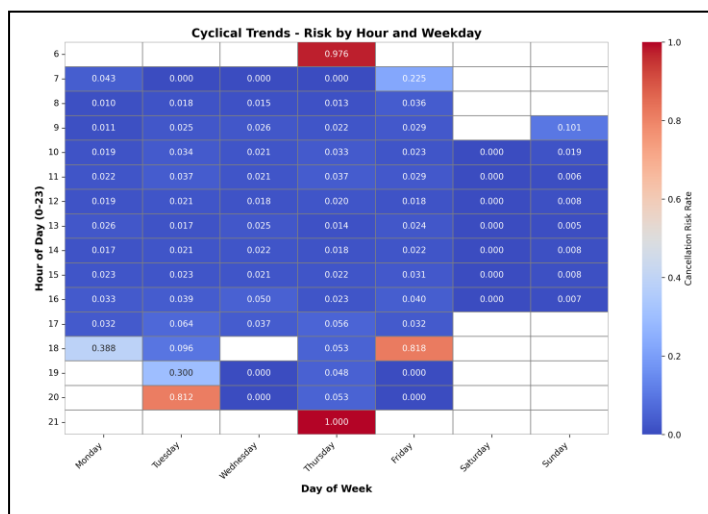


Fig. 11. Heatmap of Risk by Hour and Weekday.

8. Discussion

This paper contrasted three machine learning models in describing the risk and rate of order return or cancellation for e-commerce transactions; we identified implications for interpreting interpretable, efficient and accurate predictions within unbalanced data.

XGBoost showed the best discrimination power with ROC-AUC: 0.8852 and recall of 0.7623 for the minority class, allowed to detect high-risk transactions. The ability to capture non-linear functional relationships and interaction features supports the advantage of ensemble tree-based methods for high-dimensional imbalanced data [12]. This is consistent with previous observation that gradient boosting has the potential to trim a return-related loss by 15-20% via recall-based optimization.

The Multi Layer Perceptron had a good probability calibration ($R^2 = 0.1556$, precision = 0.8260) but a low recall (0.1152) at the score level for the minority class. This constraint doesn't work for e-commerce merchants who need to find as many risky orders as possible in both inventory management and customer retention perspectives. This precision-recall trade-off is a reflection of learning an imbalanced classification and should be tuned to the needs of the business.

Elastic Net, although interpretable, did not do well on non-linear structures in transaction data. The negative R^2 values of the Elastic Net and XGBoost learners emphasize how tough it is to shoe horn binary classification into regression in cases of extreme class imbalance. In these cases, ROC-AUC offers a more realistic discrimination evaluation compared within variance-explained metrics. The low predictive performance of Elastic Net has limited usefulness, but its interpretable coefficients provide exploratory insight on the directions/features.

SHAP value feature importance examination showed that transaction amount and product price are the most significant predictors among all models, suggesting that bigger and more expensive orders have a higher probability of being canceled possibly because of buyer or quality remorse. Geography likewise played a major role, indicating that risk assessment frameworks should take local consumer behavior differences into account.

9. Conclusion

In this study, it has been successfully concluded that the Elastic Net Regression, XGBoost and Multi-Layer Perceptron models add different (complementary) modeling powers to predict return / cancellation risks in e-commerce transactions. XGBoost performs the best for most of the practical cases and achieves a ROC-AUC score of 0.8852 with a recall rate of 0.7623, which is beneficial to early discovery on high-risk transactions. The MLP model demonstrates excellent probability calibration but further performance improvements with better precision and recall tradeoff are needed for operational deployment.

Key contributions include a strong pre-processing pipeline to handle distributions with imbalance while avoiding information leakage, thorough comparison of three different modeling strategies by means of systematic hyper-parameter optimization and explanation of risk in return or cancellation factors through use of SHAP values. These findings show that the transaction size, product price, geographic origin and time patterns work together to determine the cancellation and return behavior of customers which offers substantial insights for operational actions.

Advanced resampling methods and cost-sensitive learning need to be further studied for handling extreme class imbalance more well. Including customer reviews in unstructured data using natural language processing should help the model, as it adds information on subjective satisfaction features. Hybrid ensemble methods that utilize XGBoost in combination with neural networks might generalize even better. Furthermore, adjusting threshold on-the-fly according to business context and current operational priorities could optimize precision-recall trade-offs to suit organisational aims. By adopting these strategies, e-commerce companies could use predictive modelling to reduce economic loss and optimize supply chain management efficiency and increasing customer satisfaction rate and retention.

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